TSA 2008

# Problem Solving

Questions taken from the TSA Oxford 2008 Section 1 question paper

| Questions | Total marks | Time limit |
|---|---|---|
| 25 | 25 | 45 minutes |

### Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2008 paper, e.g. **1 (Q3)**.

### How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

**Score: _____ / 25**

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**1** (Q2) A child's bus fare is cheaper than the adult fare but is more than half the adult fare. The total cost of a single journey for an adult and two children is £1.20. Adult fares are all multiples of 10 p.

What is the adult fare?

**A** 30 p
**B** 40 p
**C** 50 p
**D** 60 p
**E** 70 p

**2** (Q6) Steven has a customised motorcycle that he is very proud of and shows regularly in competitions. It is very important to him that his presentation is perfect and he likes to stop in front of the judges with both his tyre valves in the same position. The front tyre is smaller than the rear to aid its handling. The circumference of the front wheel is 2m and the circumference of the rear wheel is 2.5m.

Steven starts with both valves in the same position at the bottom of each wheel.

How far must he drive before the valves line up again at the bottom of each wheel?

**A** 2.0 m
**B** 2.5 m
**C** 4.5 m
**D** 10.0 m
**E** 12.5 m

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**3** (Q7) I have so many four digit code numbers to remember that I cannot keep them all in my head. The banks say you should not write down your Personal Identification Numbers (PINs) for cash and credit cards and keep them in a wallet with the cards. So I can find out what the numbers are, I have devised a code which should be complicated enough to confuse any thief who steals my wallet. I code all the numbers using 1=A, 2=B up to 9=I, 0=J. Then I write the coded numbers down backwards so 1234 is written DCBA. Finally at the end of each coded number I add an extra letter between A and J to say what that particular code is of: C is for credit card; F is for phone card; B is for bicycle lock combination; E is for my Eastern Bank card and D is for my Denton Building Society card.

My list looks as follows:

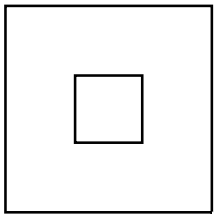
A D G E F
C J D B E
B G F A C
G J A C D
D F G C B

What is my bicycle lock combination?

**A** 4673
**B** 5741
**C** 7613
**D** 4356
**E** 3764

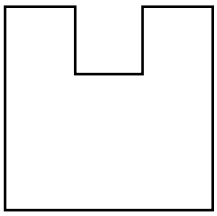
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**4** (Q8) The drawing shows a sculpture as seen from above.

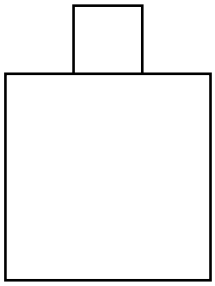


Which of these could **not** be a view from one of the sides?

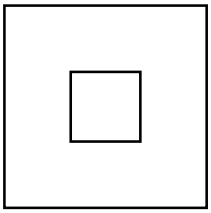
**A**



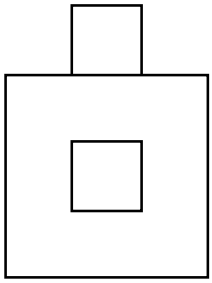
**B**



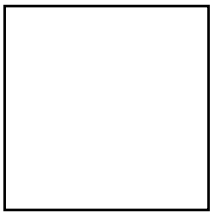
**C**



**D**



**E**



**5** (Q12) Pierre and Marc are waiters. In one particular month, Pierre worked 30 sessions at the normal rate and 10 sessions at the overtime rate. Marc worked 20 sessions at the normal rate and 5 sessions at the overtime rate. At the end of this month, Pierre earned 700 euros. Marc earned 425 euros.

What is the overtime rate per session?

**A** 35 euros

**B** 30 euros

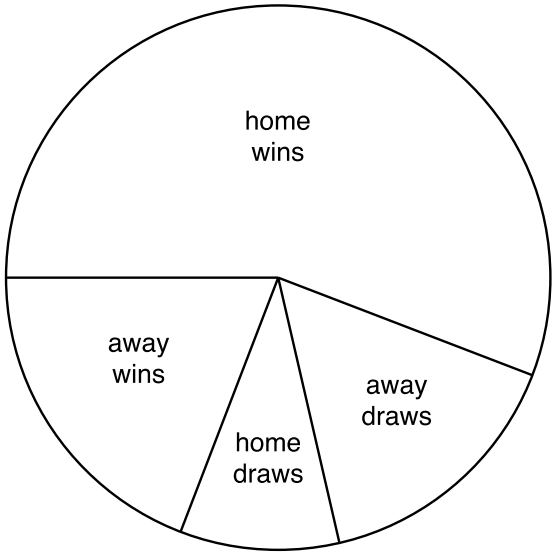
**C** 25 euros

**D** 20 euros

**E** 15 euros

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**6** (Q13) The chart below shows the distribution of results this season for the Arlsea Strikers hockey team. They have played 24 matches to date and lost none.

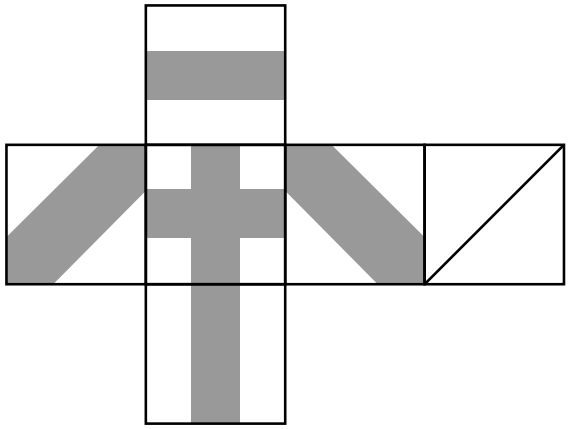



How many matches have they won?

**A** 4
**B** 6
**C** 14
**D** 16
**E** 18

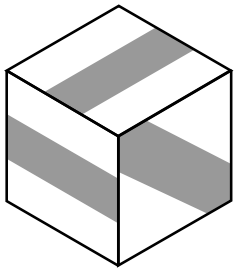
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**7** (Q14) The cut out below was folded together to make a cube.

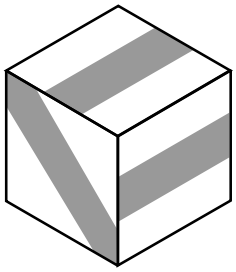


Which of the following cubes was formed from this cut-out?

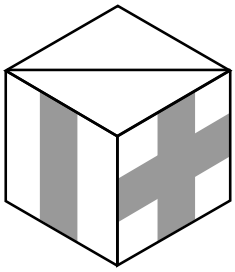
**A**



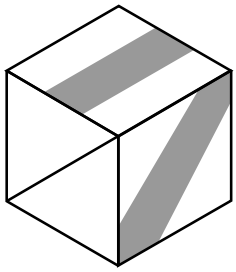
**B**



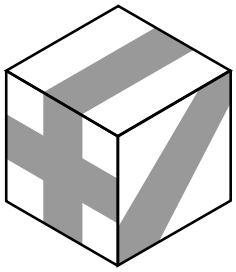
**C**



**D**



**E**



**8** (Q18) Inflation in Bolandia has been a steady 10% per year for many years. Evitan, a Bolandian citizen, bought his car one year ago for 500 Bols. He now wants to trade in his car for an identical model. He can expect to receive 80% of the current new value of his existing car. The price of new models has increased in line with inflation.

How many extra Bols will he have to pay for his new car?

**A** 50
**B** 55
**C** 60
**D** 100
**E** 110

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**9** (Q19) The table below shows the unemployment rate and percent change in industrial production over the last year for several countries.

| | ***INDUSTRIAL PRODUCTION*** Annual Change % | | ***UNEMPLOYMENT*** Rate % | |
|---|---|---|---|---|
| | Latest | Year ago | Latest | Year ago |
| Belgium | −2.6 | −2.5 | 8.9 | 10.6 |
| Canada | 6.9 | 5.4 | 9.6 | 11.2 |
| France | 4.7 | −4.3 | 12.7 | 12.1 |
| Germany | 5.5 | −4.0 | 9.1 | 9.0 |
| Italy | 7.3 | −1.3 | 12.5 | 10.7 |
| Japan | 5.9 | −3.2 | 2.9 | 2.8 |
| Netherlands | 1.2 | 2.4 | 9.6 | 8.4 |
| Spain | 4.9 | −6.8 | 24.1 | 22.1 |
| Sweden | 8.2 | −1.8 | 8.1 | 8.7 |
| UK | 4.2 | 3.3 | 8.8 | 10.0 |
| USA | 5.7 | 3.4 | 5.4 | 6.4 |

Which country has had a larger percentage increase in industrial production and greater fall in the percentage unemployed than the UK over the last year?

**A** Belgium

**B** Canada

**C** France

**D** Italy

**E** USA

**10** (Q20) The roads on Bolandia all run either north - south or east - west and are laid out on a 5km grid. Ahmed and Wayne are delivery drivers. One morning, Ahmed starts at Abbeytown and drives 5km west then 15km south to reach Barneyville. He then drives 5km east to Carloston. Wayne started at Barneyville, drove 5km south to Denburgh then 10km east and 15km north to Easterby. Ahmed then rings Wayne to arrange to meet for lunch.

If one of them stays where he is, what is the minimum distance the other must drive to meet him?

**A** 10 km
**B** 15 km
**C** 25 km
**D** 30 km
**E** 35 km

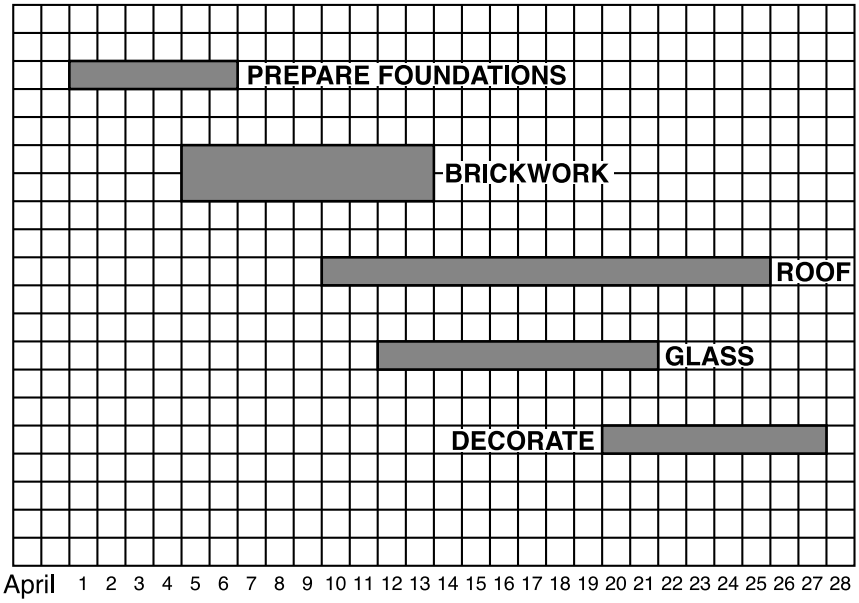
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**11** (Q24) A long-running play in London's West End is performed every evening except Sunday. In addition, there are matinee performances every Wednesday, Thursday and Saturday.

What is the maximum number of performances that can be given during a calendar month of 31 days?

**A** 36
**B** 38
**C** 39
**D** 41
**E** 42

**12** (Q25) Peter is having a large conservatory built at the back of his house. Below is a plan of the project, showing when each part of the conservatory is to be built.




Each blocked square on the plan represents one man for one day.

How many days labour will Peter have to pay for, and what is the maximum number of workers on site at the same time?

**A** 48 and 3
**B** 48 and 4
**C** 58 and 3
**D** 58 and 4
**E** 58 and 6

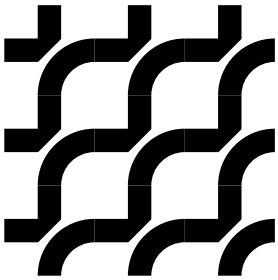
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**13** (Q26) The design of a tile for a bathroom wall is shown below:

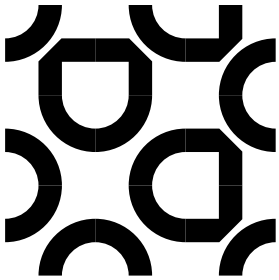


Which of the following patterns may **NOT** be made using only a number of identical tiles as shown?

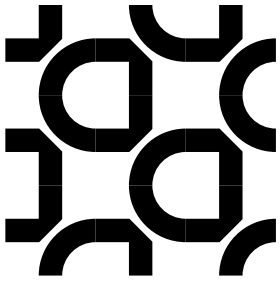
**A**



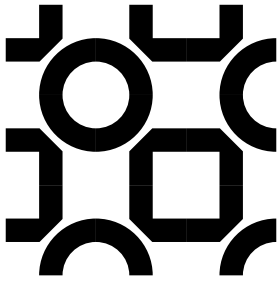
**B**



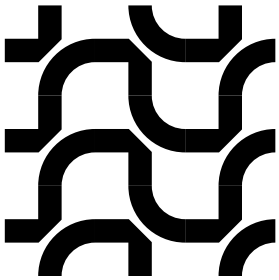
**C**



**D**

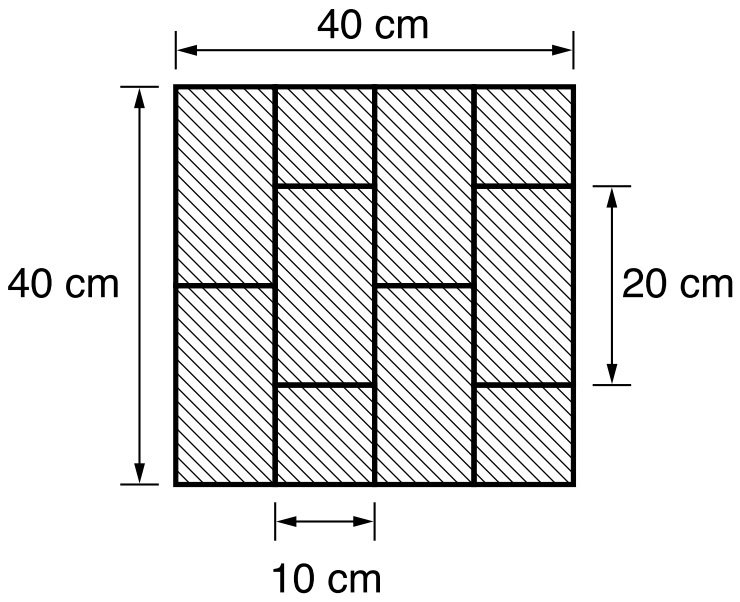


**E**



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**14** (Q30) I am resurfacing my driveway using bricks in the pattern shown below. The visible part of the brick measures 10cm by 20cm. To achieve a rectangular shape, part of some of the bricks must be cut off. The part which is cut off can be used elsewhere on the drive. The diagram shows that eight bricks would be required to cover an area 40cm by 40cm:




What is the minimum number of bricks needed to resurface a rectangular drive measuring 550cm by 400cm?

**A** 1080 bricks
**B** 1100 bricks
**C** 1120 bricks
**D** 1127 bricks
**E** 1128 bricks

**15** (Q31) John has 15kg of general potting compost which is made up of 1/3 sand and 2/3 coir. He wants compost to repot some conifers which require a better draining mixture of 60% sand to 40% coir.

How much sand does he have to add to his general purpose compost to make it suitable for conifers?

**A** 5 kg
**B** 10 kg
**C** 15 kg
**D** 20 kg
**E** 25 kg

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**16** (Q32) A netball tournament is divided into two stages. In the first stage, the teams are divided into equal groups within which each team plays every other team once. In the second stage, the winners of each group take part in a knockout competition which starts with four quarter finals.

Which of the following could be the number of teams starting the tournament?

**A** 8
**B** 20
**C** 44
**D** 50
**E** 72

**17** (Q36) Below is part of an itinerary for flying from London to Kampala. It involves changing flights at Dubai with a wait of 5 hours 15 minutes. All times are local times. Dubai is 4 hours ahead of London and Kampala is 1 hour behind Dubai.

| | |
|---|---|
| DEPART LONDON (HEATHROW) | 22:30 FRI 30.01.04 |
| DEPART DUBAI | 14:30 SAT 31.01.04 |
| ARRIVE KAMPALA (ENTEBE) | 20:45 SAT 31.01.04 |

What is the total time spent in the air?

**A** 12 hrs 30 mins
**B** 14 hrs 00 mins
**C** 15 hrs 00 mins
**D** 17 hrs 00 mins
**E** 19 hrs 15 mins

**18** (Q37) Peter needs to separate his sheep into 16 square pens. Each side of a pen is formed by a free-standing barrier known as a hurdle. Each pen must be individually accessible on at least one side so that sheep can be penned or released without a risk of the others escaping.

What is the smallest number of hurdles that Peter must take out to the fields in order to accomplish his task?

**A** 40
**B** 42
**C** 48
**D** 49
**E** 64

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**19** (Q38) A piece of ribbon 8m long is folded in half so the two ends are on top of each other. This doubled 4m length is then folded in half again. The folded, 2m length of ribbon is then cut right through at its midway point.

How many cut pieces are there and what are their lengths?

**A** 8 pieces of ribbon each 1m long.

**B** 6 pieces of ribbon, 2 of 2m long and 4 of 1m long.

**C** 5 pieces of ribbon, 3 of 2m long and 2 of 1m long.

**D** 4 pieces of ribbon each 2m long.

**E** 3 pieces of ribbon one 4m long and 2 of 2m long.

**20** (Q42) A small mail order company sends out three sizes of parcel in about equal numbers which require postages of 24c, 30c and 72c. They stock stamps in denominations of 6c and 30c and use the minimum number of stamps on each parcel.

What proportion of the two denominations of stamps should the company buy?

**A** 1 x 6c for every 4 x 30c

**B** 1 x 6c for every 1 x 30c

**C** 2 x 6c for every 1 x 30c

**D** 5 x 6c for every 1 x 30c

**E** 11 x 6c for every 2 x 30c

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**21** (Q43) Every year the 12 local teams that compete for the Tipley and District Tiddlywinks Trophy are split into two leagues for the purpose of eliminating four of the teams, and sorting out the pairings for the knockout stage of the competition. The table below shows how this is done (QF is Quarter Final).

| | |
|---|---|
| 1st QF | Winners League A v 4th place League B |
| 2nd QF | 3rd place League A v Runners-up League B |
| 3rd QF | Winners League B v 4th place League A |
| 4th QF | 3rd place League B v Runners-up League A |
| 1st Semi-final | Winners of 1st QF v Winners of 2nd QF |
| 2nd Semi-final | Winners of 3rd QF v Winners of 4th QF |

This year's league stage has just been completed, with the following results.

| | League A | League B |
|---|---|---|
| 1st | The Scorpions | The Scales |
| 2nd | The Fish | The Bulls |
| 3rd | The Bearers | The Goats |
| 4th | The Rams | The Archers |
| 5th | The Crabs | The Maidens |
| 6th | The Lions | The Twins |

Despite having finished 3rd in their league, the reigning champions, The Goats are still confident of reaching the final.

If The Goats are successful, what are the alternatives as to their prospective opponents in the final?

**A** The Scorpions or The Fish or The Bearers or The Rams.

**B** The Archers or The Bulls or The Rams or The Fish.

**C** The Scorpions or The Archers or The Bearers or The Bulls.

**D** The Scorpions or The Archers or The Scales or The Rams.

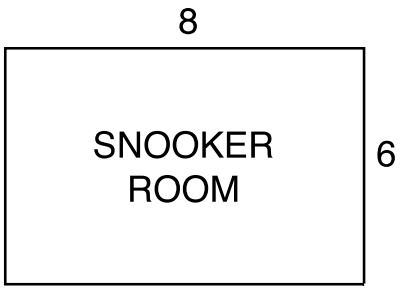
**E** The Scorpions or The Twins or The Bearers or The Bulls.

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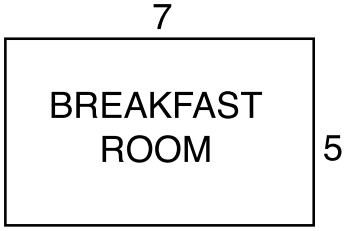
**22** (Q44) Jess buys two lengths of carpet in the sales. The carpet is 4m wide and one of the lengths purchased is 4m long while the other is 6m. There is a very pronounced pattern which means that the pieces can only be joined so that the pattern runs the same way. Jess will only accept one join.

In which of Jess's rooms could the carpet be fitted?

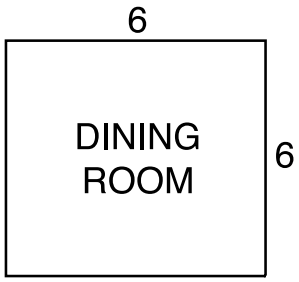
**A**



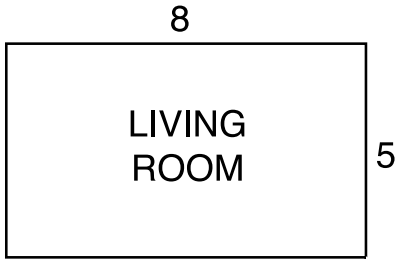

**B**



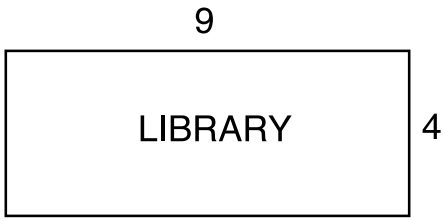

**C**




**D**




**E**




**23** (Q48) In a prize weight-lifting competition the winner receives $100 if he lifts 80kg. He then receives $10 for the next 5 additional kg, $15 for the second, and the incremental prize money increases by a further $5 for each additional unit of 5kg he successfully lifts.

The winner of the competition lifts 110kg.

How much prize money does he receive?

**A** $ 130

**B** $ 160

**C** $ 235

**D** $ 240

**E** $ 310

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**24** (Q49) The table below shows the expenses by month for a given household.

| | *Jan* | *Feb* | *Mar* | *Apr* | *May* | *Jun* | *Jul* | *Aug* | *Sep* | *Oct* | *Nov* | *Dec* | *Total* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *Expenses* | | | | | | | | | | | | | |
| Gas | 125 | | | 120 | | | 80 | | | 70 | | | 395 |
| Electricity | | 160 | | | 140 | | | 80 | | | 110 | | 490 |
| Telephone | | | 80 | | | 95 | | | 110 | | | 90 | 375 |
| Council Tax | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 60 | 720 |
| Coal | | | | | | 80 | | | | | | 85 | 165 |
| Road Tax | | | | | | | | | | 125 | | | 125 |
| Car Insurance | | | | | | | | | | 250 | | | 250 |
| Car Maintenance | | 125 | | | | | | 155 | | | 285 | | 565 |
| TV Licence | | | | | | | | | 85 | | | | 85 |
| Logs for Fire | | | 40 | | 40 | | 60 | | | | | | 140 |
| House Maintenance | | | 55 | | | 225 | | | | | 128 | | 408 |
| Disposable Items | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 10 | 120 |
| Food | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 200 | 2400 |
| Mortgage | 320 | 320 | 320 | 320 | 320 | 320 | 320 | 320 | 320 | 320 | 320 | 320 | 3840 |
| House Insurance | 240 | | | | | | | | | | | | 240 |
| *Total Expenses* | 955 | 875 | 765 | 710 | 770 | 990 | 730 | 825 | 785 | 1035 | 1113 | 765 | 10318 |

In which month were the energy costs highest?

**A** January

**B** February

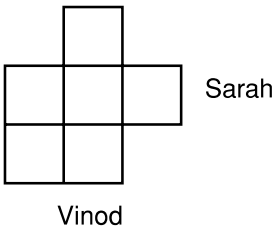
**C** April

**D** May

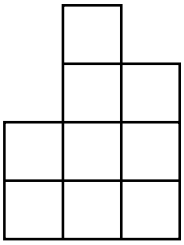
**E** July

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**25** (Q50) Vinod and Sarah are about to move some cube-shaped boxes that have been left in the middle of a store room. The appearance of the boxes from above and the positions of Vinod and Sarah are shown below.

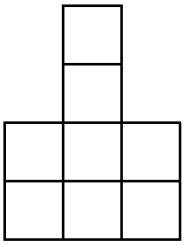



Vinod sees this:

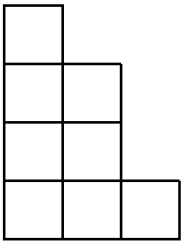


Which of the following **cannot** be the view that Sarah sees?

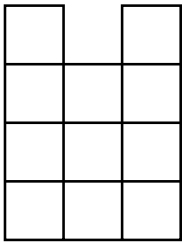
**A**



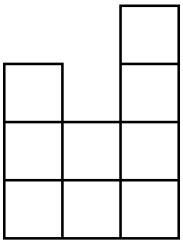
**B**



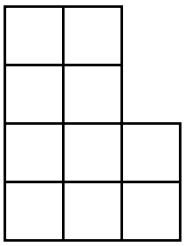
**C**



**D**



**E**



**END OF TEST**

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